

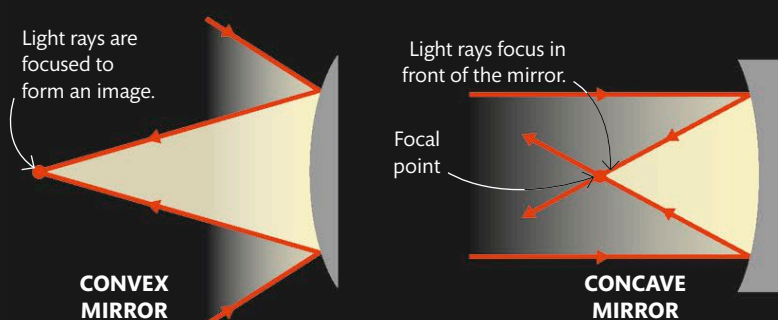
SCANNING THE SKIES

TELESCOPES

For centuries, humans have looked at the skies and studied objects in space. But the invention and development of the telescope has enabled us to look farther into the Universe, making distant objects ever more visible. In a telescope, light from a distant object such as a comet, planet, or star is gathered and brought into focus, forming a bright image, which can then be magnified. There are different types of telescopes—reflecting telescopes use curved mirrors and refracting telescopes use lenses. Some telescopes detect other types of electromagnetic radiation, such as radio waves.

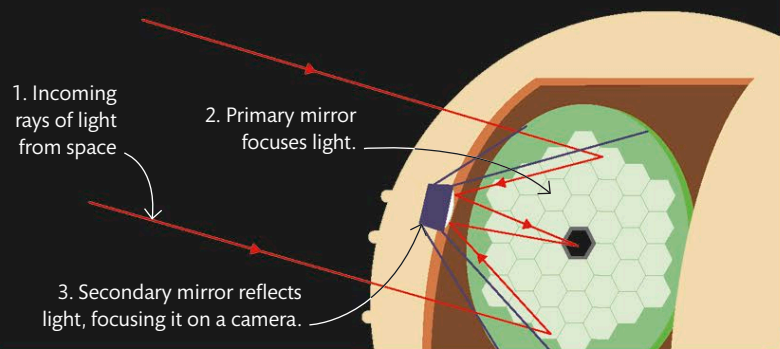
TELESCOPE MIRRORS

Light rays, or waves, reflect, or bounce off, objects (see page 70). Mirrors reflect nearly all the light that hits them, and curved mirrors can be used to focus light to a point. A concave mirror is curved inward like a bowl, while a convex mirror is curved outward like a dome.



How reflecting telescopes work

A reflecting telescope contains a large, concave primary mirror and a smaller secondary mirror. The primary mirror receives the incoming light from an object in space and focuses it on the secondary mirror. The secondary mirror then reflects this light toward a camera, allowing the telescope to take pictures of the distant object.



Eye on space

Perched on top of an extinct volcano in Spain's Canary Islands, the Gran Telescopio Canarias—the world's largest reflecting telescope—is located far away from pollution and city lights, making it ideal for observing objects in space in great detail. Its primary mirror is made up of 36 hexagonal segments, which collectively behave like a single giant mirror 34 ft (10.4 m) in diameter.

